



**Introduction to
Internet of Things
Assignment-Week 6**

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15

Total marks: 15 X 1= 15

QUESTION 1:

What are the five basic data types available in Python, as introduced in the lecture?

- a. Numbers, Strings, List, Tuple, Dictionary
- b. Integer, Float, Boolean, Char, Byte
- c. Array, Struct, Class, Object, Set
- d. Numbers, Strings, Queue, Stack, Graph

Correct Answer: a. Numbers, Strings, List, Tuple, Dictionary

Detailed Solution: The lecture states that Python has five data types: Numbers, String, List, Tuple, and Dictionary.

See Lecture 26 (Introduction to Python Programming – Part I) @ 15:50

QUESTION 2:

Which keyword is used to define a function in Python?

- a. func
- b. def
- c. define
- d. function

Correct Answer: b. def

Detailed Solution: A function in Python is defined using the def keyword, followed by the function name and its arguments.

See Lecture 26 (Introduction to Python Programming – Part I) @ 19:20



QUESTION 3:

In Python's exception handling, which keyword is used to handle an exception raised inside the try block?

- a. catch
- b. except
- c. handle
- d. rescue

Correct Answer: b. except

Detailed Solution: Python uses the try, except and else block structure for exception handling; the except block catches the exception raised in the try block.

See Lecture 26 (Introduction to Python Programming – Part I) @ 28:55

QUESTION 4:

Which mode of Python's open() function is used to add data to the end of an existing file without erasing its existing content?

- a. r (read mode)
- b. w (write mode)
- c. a (append mode)
- d. r+ (read and write mode)

Correct Answer: c. a (append mode)

Detailed Solution: Python supports four basic file modes: r (read), w (write), a (append), and r+ (read and write). The append mode adds new data to the end of the file without deleting existing content.

See Lecture 27 (Introduction to Python Programming – Part II) @ 02:55

QUESTION 5:



Which Python library, installable via pip, is used for image read/write operations as covered in the lecture?

- a. NumPy
- b. PIL / Pillow
- c. OpenCV
- d. Matplotlib

Correct Answer: b. PIL / Pillow

Detailed Solution: PIL (Python Image Library) is used to work with image files. It is installed using 'sudo pip install pillow'; PIL supports up to Python 2.7 while Pillow supports Python 3.x.

See Lecture 27 (Introduction to Python Programming – Part II) @ 17:25

QUESTION 6:

Which socket_type value is used while creating a TCP/IP socket in Python's socket module?

- a. SOCK_DGRAM
- b. SOCK_STREAM
- c. SOCK_RAW
- d. SOCK_SEQPACKET

Correct Answer: b. SOCK_STREAM

Detailed Solution: A socket is created using `socket.socket(socket_family, socket_type, protocol)`. For a TCP/IP (connection-oriented) socket, `socket_family` is `AF_INET` and `socket_type` is `SOCK_STREAM`.

See Lecture 27 (Introduction to Python Programming – Part II) @ 24:40

QUESTION 7:

How many GPIO pins does the Raspberry Pi 3 Model B have, as shown in the specifications table?

- a. 26
- b. 32



c. 40

d. 44

Correct Answer: c. 40

Detailed Solution: As per the specifications table, the Raspberry Pi 3 Model B, Raspberry Pi 2 Model B, and Raspberry Pi Zero all have 40 GPIO pins.

See Lecture 28 (Introduction to Raspberry Pi – Part I) @ 05:10

QUESTION 8:

Which of the following is an officially supported operating system for Raspberry Pi mentioned in the lecture?

a. Raspbian

b. Windows XP

c. macOS

d. Solaris

Correct Answer: a. Raspbian

Detailed Solution: Raspbian and NOOBS are the officially supported operating systems for Raspberry Pi. Ubuntu Mate, Snappy Ubuntu Core, Windows 10 Core, Pinet, and RISC OS are examples of third-party operating systems.

See Lecture 28 (Introduction to Raspberry Pi – Part I) @ 12:05

QUESTION 9:

Raspberry Pi is best described as a:

a. Printer

b. Smartphone

c. Wi-Fi router

d. Single-board computer

Correct Answer: d. Single-board computer



Detailed Solution: The lecture introduces Raspberry Pi as a computer that fits in your palm — a low-cost, single-board computer that is easy to access and use.

See Lecture 28 (Introduction to Raspberry Pi – Part I) @ 04:45

QUESTION 10:

Which command is used to install the GPIO library on Raspberry Pi before programming the blinking LED example?

- a. `sudo apt-get install python-rpi.gpio`
- b. `pip install gpio`
- c. `sudo apt-get install led-lib`
- d. `npm install rpi-gpio`

Correct Answer: a. `sudo apt-get install python-rpi.gpio`

Detailed Solution: After installing Python development tools using '`sudo apt-get install python-dev`', the GPIO library is installed using '`sudo apt-get install python-rpi.gpio`'.

See Lecture 29 (Introduction to Raspberry Pi – Part II) @ 07:43

QUESTION 11:

Where is the CSI camera connector slot located on the Raspberry Pi board?

- a. Between the USB ports
- b. Between the Ethernet port and the HDMI port
- c. Next to the power supply pin
- d. On the underside of the board

Correct Answer: b. Between the Ethernet port and the HDMI port

Detailed Solution: The Raspberry Pi Camera connects through a dedicated CSI slot, which is placed between the Ethernet port and the HDMI port on the board.

See Lecture 29 (Introduction to Raspberry Pi – Part II) @ 18:27



QUESTION 12:

Which terminal command captures a still image using the Raspberry Pi camera?

- a. raspistill -o image.jpg
- b. raspicam capture
- c. pi-camera snap
- d. capture.py --img

Correct Answer: a. raspistill -o image.jpg

Detailed Solution: Running 'raspistill -o image.jpg' in the terminal captures an image using the Raspberry Pi camera and stores it as image.jpg.

See Lecture 29 (Introduction to Raspberry Pi – Part II) @ 19:35

QUESTION 13:

What does DHT stand for in the DHT sensor used with Raspberry Pi in the auto-cooling system?

- a. Digital Heat Transfer
- b. Digital Humidity and Temperature
- c. Direct Hardware Trigger
- d. Dual Humidity Thermostat

Correct Answer: b. Digital Humidity and Temperature

Detailed Solution: DHT stands for Digital Humidity and Temperature Sensor. It has 4 pins: PIN 1 (3.3V–5V power supply), PIN 2 (Data), PIN 3 (Null), and PIN 4 (Ground).

See Lecture 30 (Implementation of IoT with Raspberry Pi – Part I) @ 05:20

QUESTION 14:

How many output terminals does the relay used in the temperature-dependent auto-cooling system have, as shown in the lecture?



- a. 2
- b. 3
- c. 4
- d. 5

Correct Answer: b. 3

Detailed Solution: The relay is a mechanical/electromechanical switch with 3 output terminals (left to right): NO (normal open), Common, and NC (normal close).

See Lecture 30 (Implementation of IoT with Raspberry Pi – Part I) @ 07:50

QUESTION 15:

State whether the following statement is True or False.

Statement: In the temperature-dependent auto-cooling system, the fan operates when the surrounding temperature rises above the threshold value specified in the code.

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: The lecture demonstrates that the fan is switched on whenever the measured temperature exceeds the threshold value set in the program.

See Lecture 30 (Implementation of IoT with Raspberry Pi – Part I) @ 18:14

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